

HOW TO MAKE 3D CABs

Includes a 3D Cab

A Quick Intro Guide

FOR OPEN RAILS WITH BLENDER 3



By ExRail

GETTING READY

Download blender 3.5 from
WWW.BLENDER.ORG
And learn to use it a bit!

Download S file Exporter plugin by
by Wayne Campbell
SITES.GOOGLE.COM/VIEW/BLENDERTOMSTS

SUPPORT TOOLS

Notepad++ (CVF 3D cab)
NOTEPAD-PLUS-PLUS.ORG

Gimp 2.10 (DDS support)
WWW.GIMP.ORG

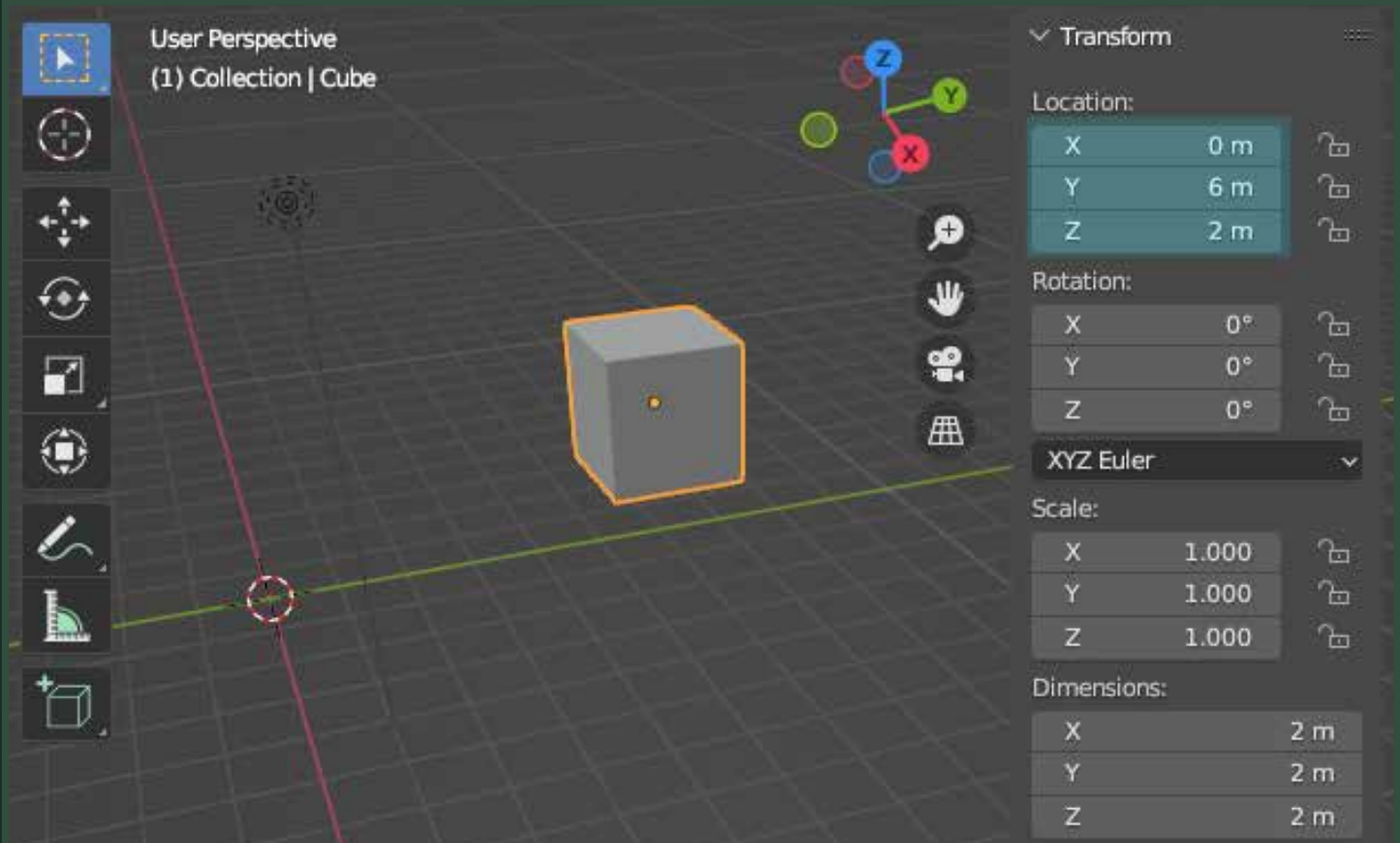
Total Commander
WWW.GHISLER.COM

CONTENT

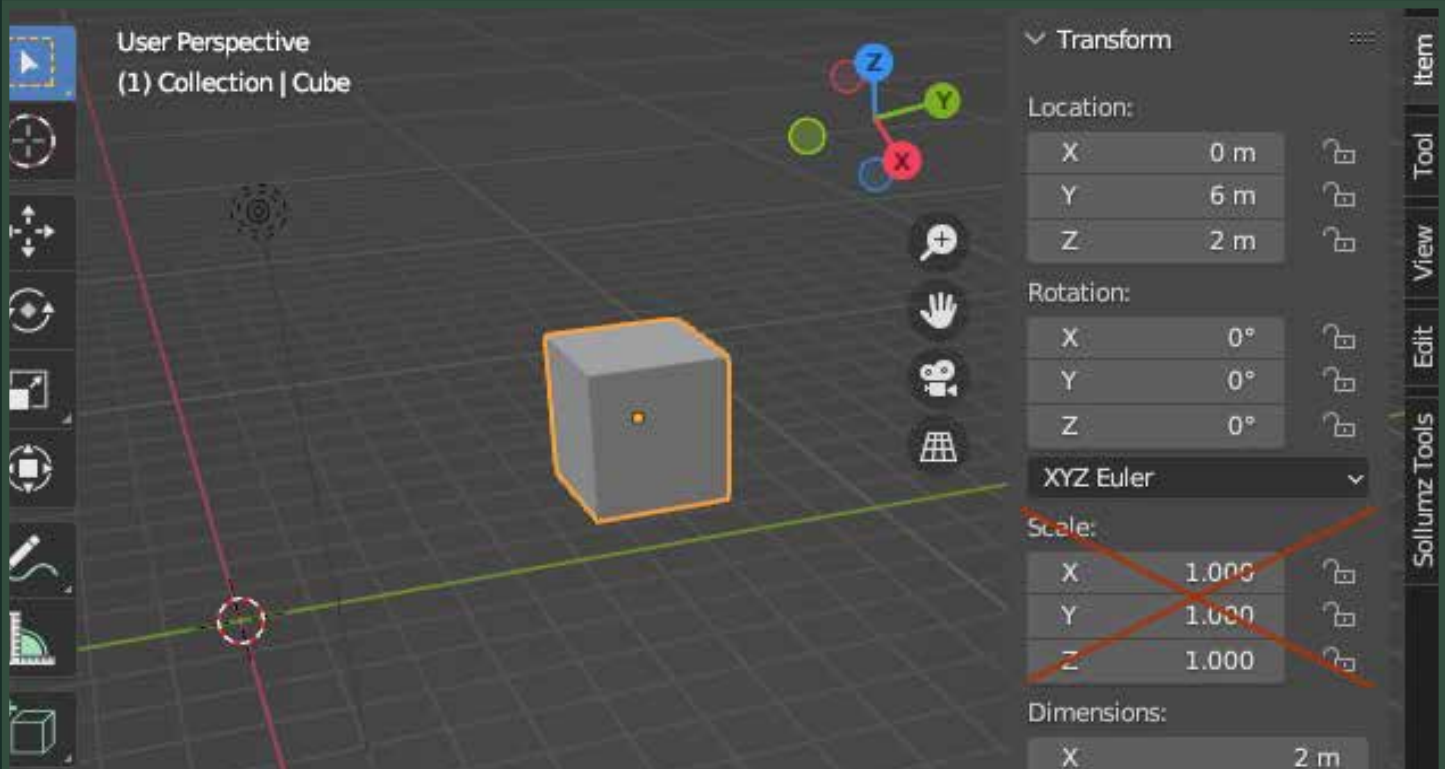
STARTING FROM SCRATCH A QUICK GUIDE IN 9 PAGES

1. Placing The Default Cube
2. Adding a plane
3. Changing the dimensions
4. Material & Illumination
5. Collection and Instrument Names
6. The 3D cab .cvf file
7. The 3D cab & the eng file
8. Export .s file
9. Testing

PLACING THE DEFAULT CUBE



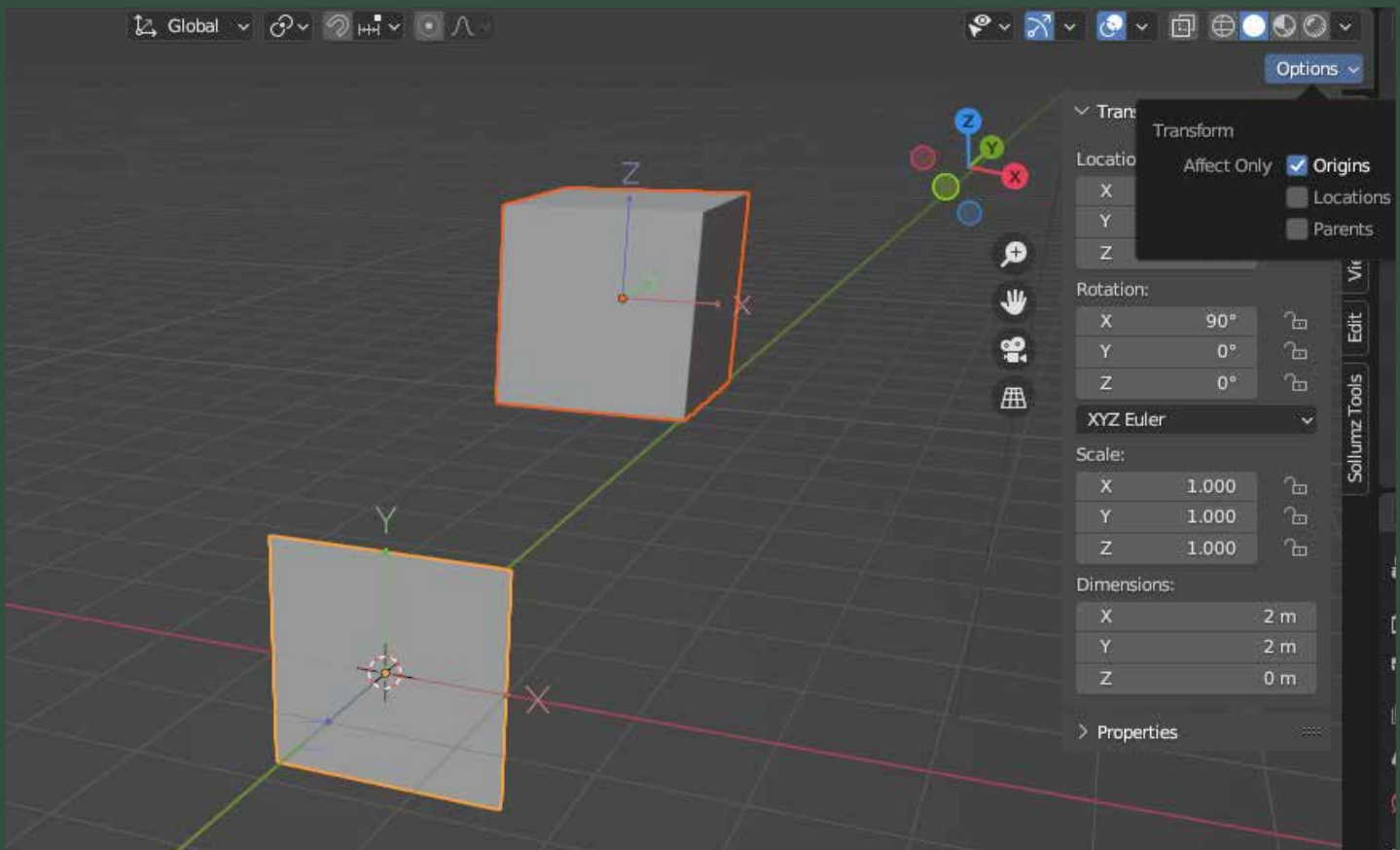
A good place to start is properly around $X=0, Y=6, Z=2$, change the numbers to move it there.



No scaling for objects to be used as instruments !

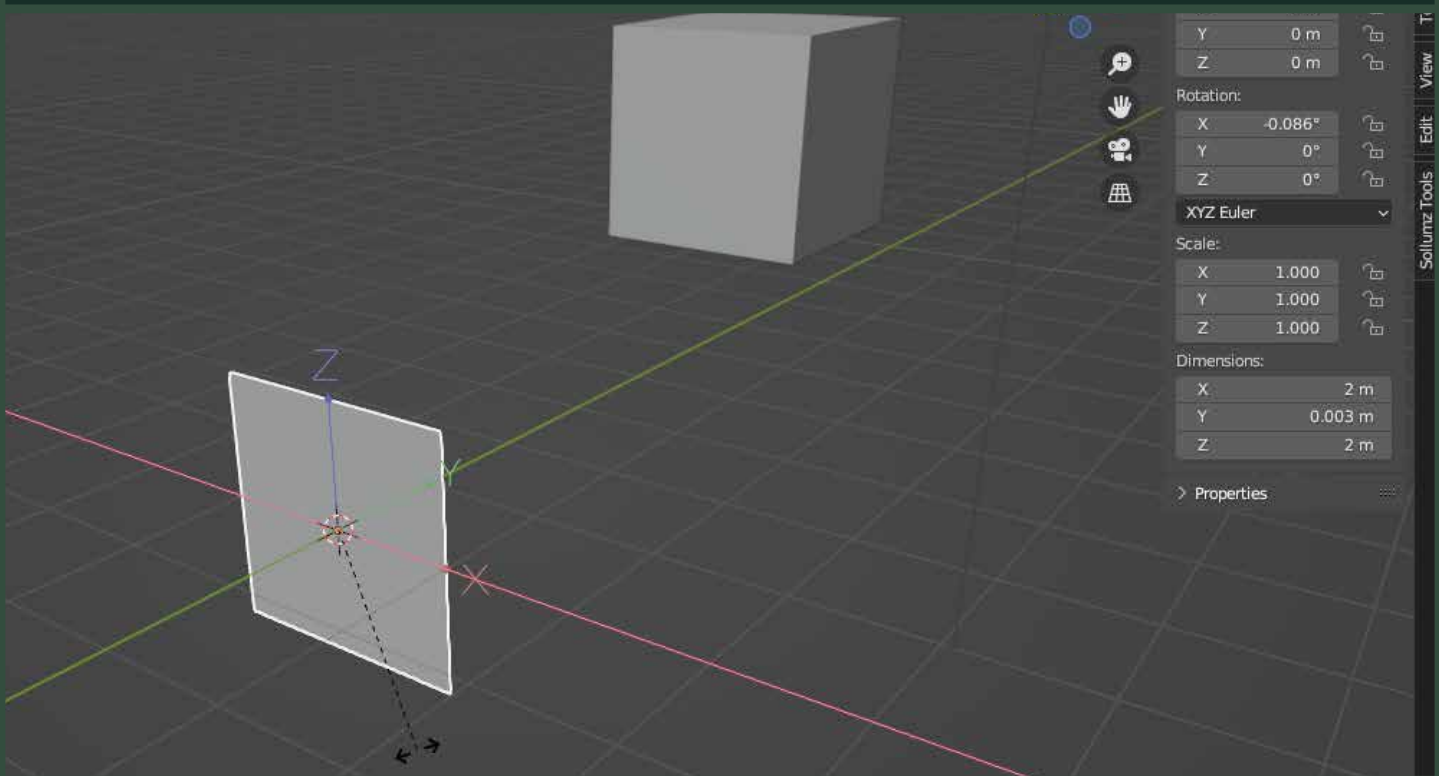
All instruments must be made with a scale 1.0 or they will appear bigger or smaller in Open Rails 3D cab, scaling can be done in Edit Mode.

ADDING A PLANE

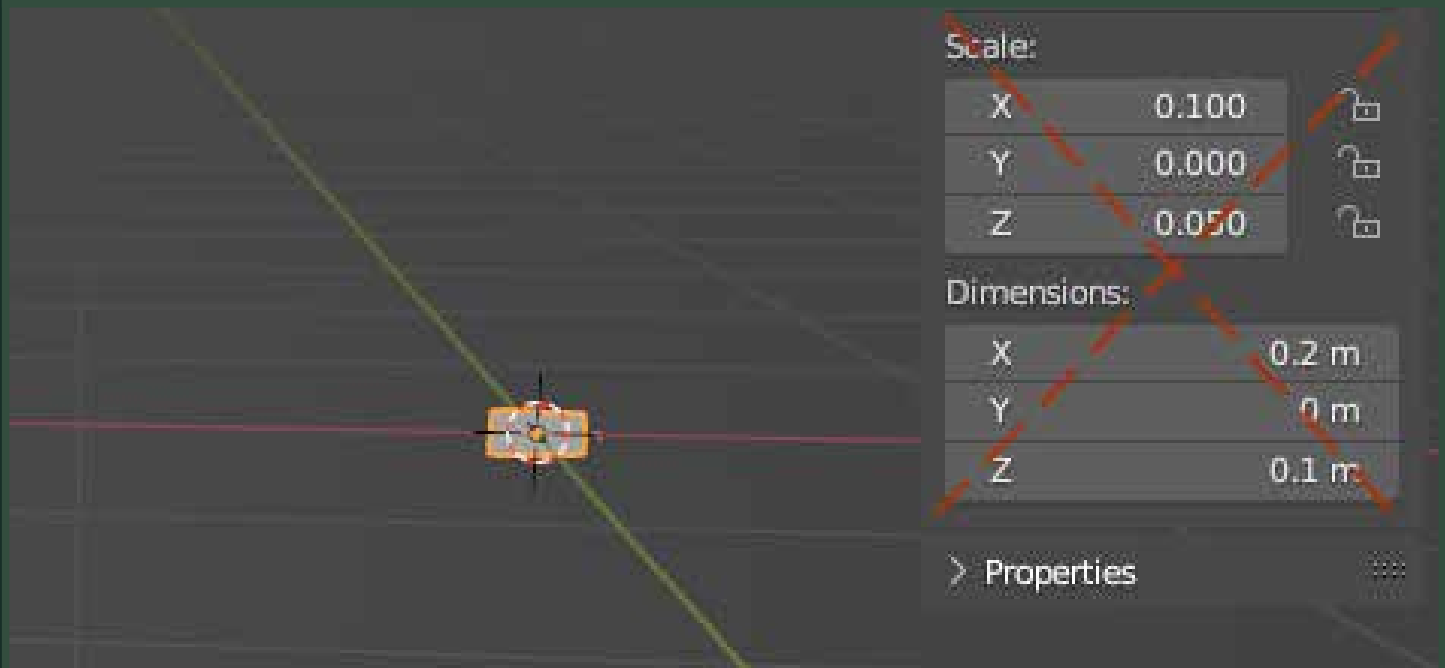


Add a plane and **Rotate it Into X=90°** to be used as a Digital clock.
Note that the orientation of the pivot point is wrong! **Enable 'Affect Only'**

Rotate the pivot point 90° to match the cubes so **Z is up, Y forward** and Remember To Disable 'Effect Only' again

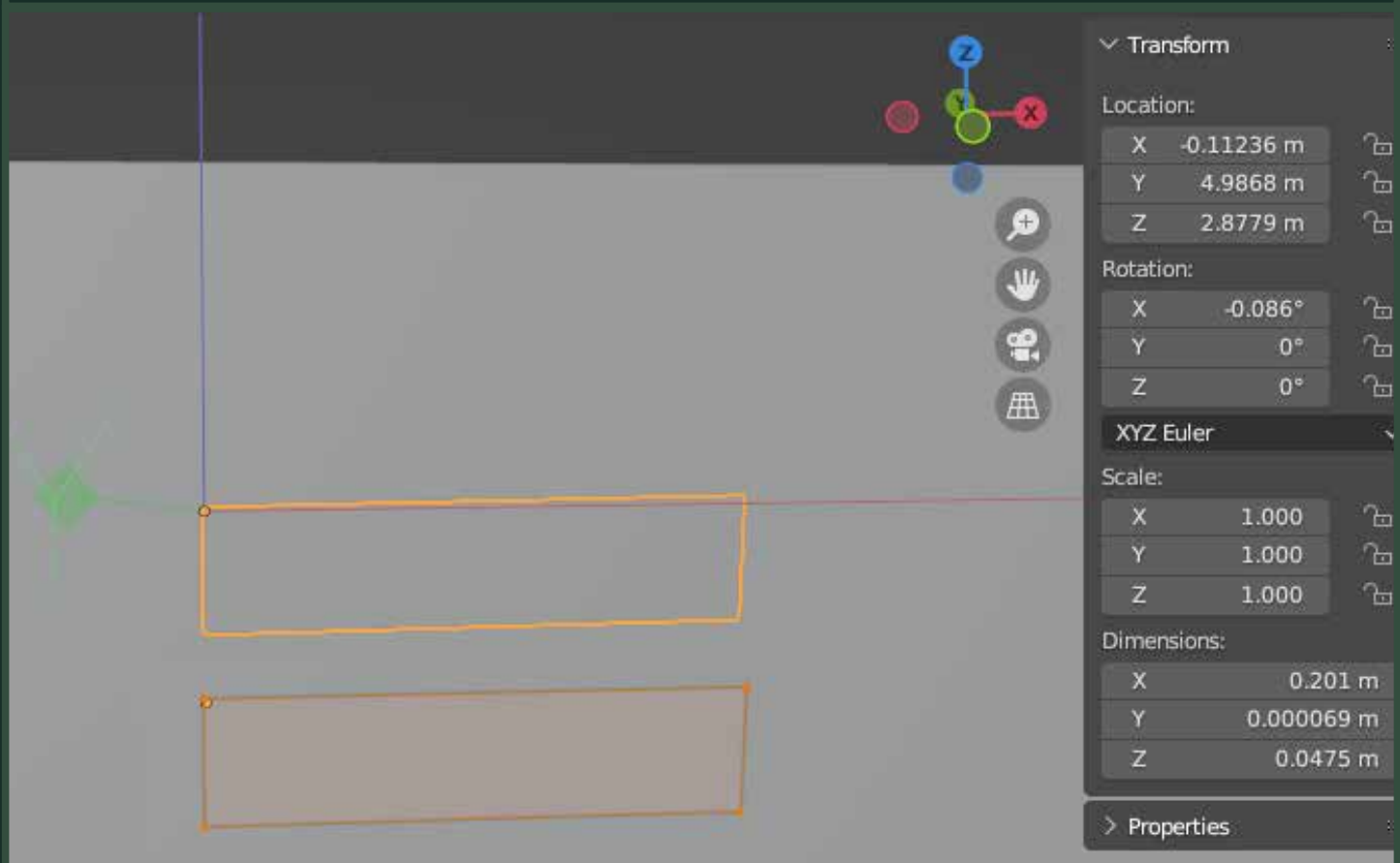


CHANGING THE PLANES DIMENSIONS



In Object mode altering the Dimensions also effect the scaling. **CTRL+A Scale set it to 1.0**

NB: Much time can be spend on fault finding where it turns out to be caused by scaling or pivot orientation.



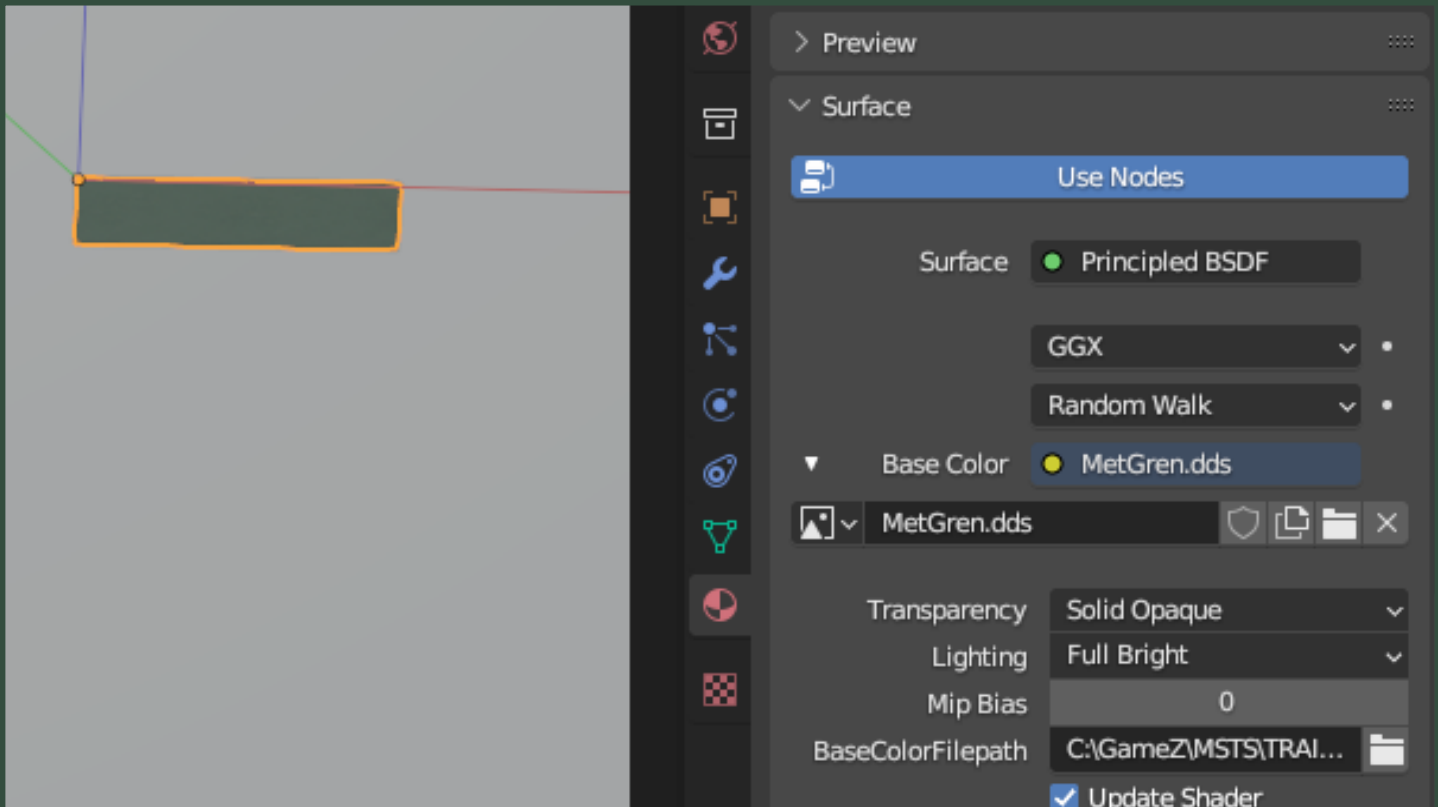
Go into Edit Mode

Change the vertex positions to get a **20 cm wide** and **5 cm high** rectangle, either by scaling or movement.

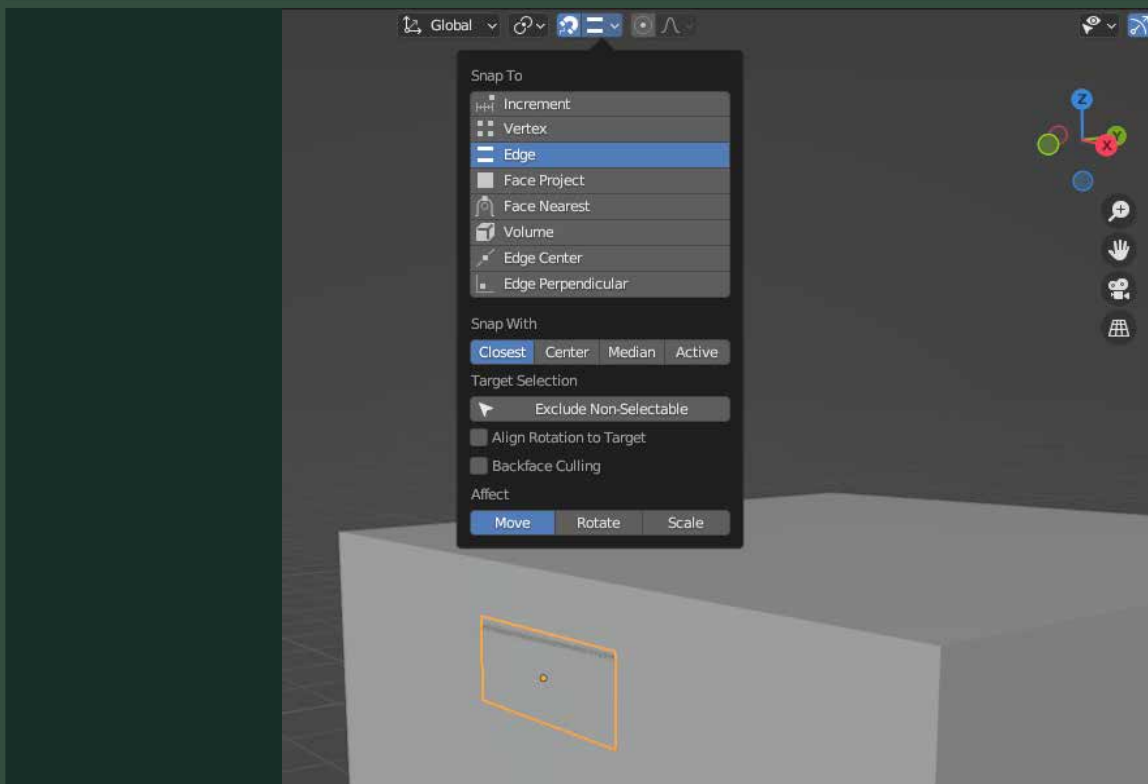
Go into Object Mode

Enable 'Affect Only' and move the pivot point to the top left corner **Disable 'Affect Only'**

MATERIAL

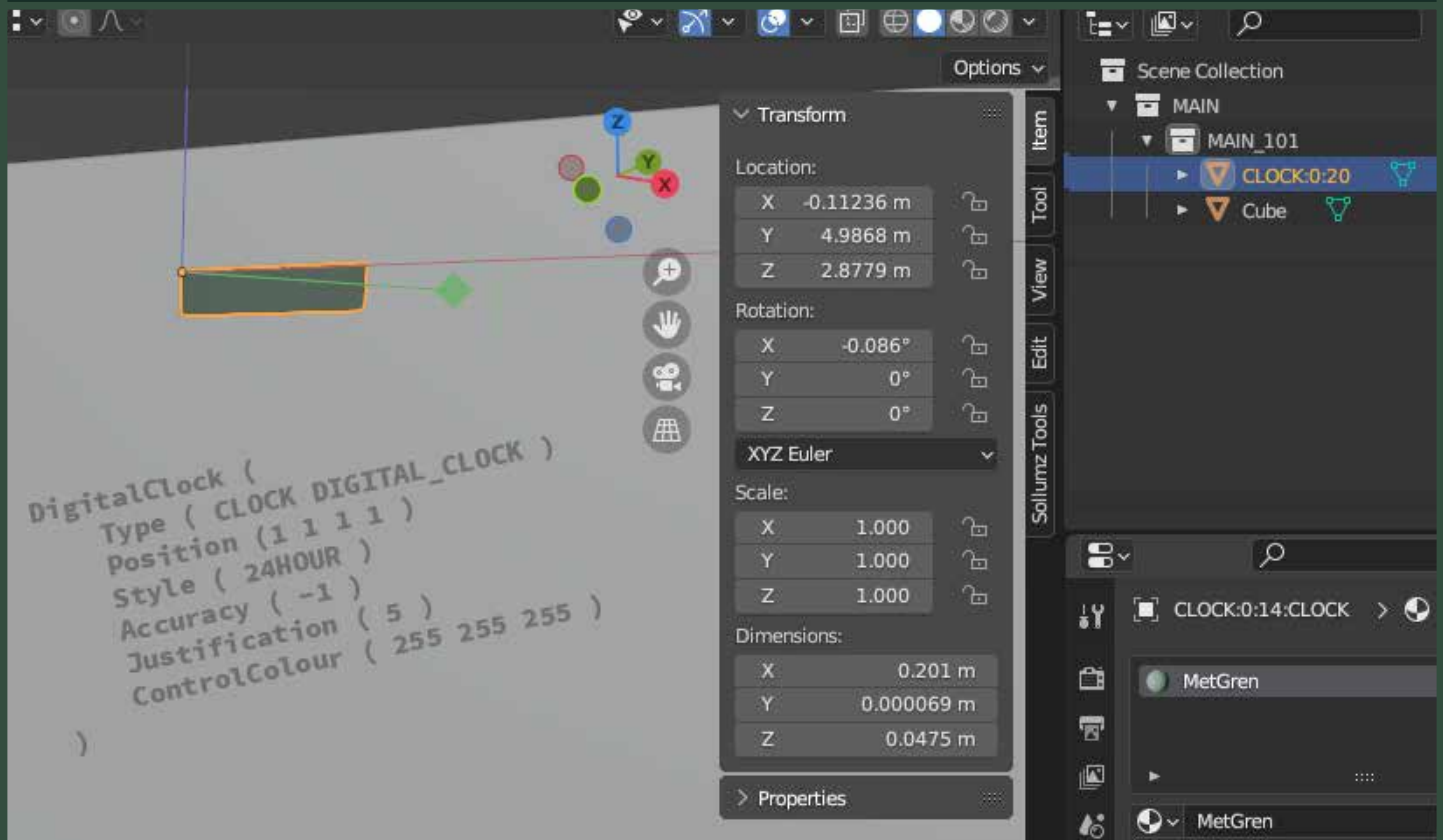


Create and assign a image texture material and set the illumination to **Full Bright** and point to the texture.



Enable **Snap to Edge** and move the plane close to the cube to snap it. Note the flickering Z collision, move it a bit out from the cube to avoid that.

COLLECTION AND INSTRUMENT NAMING



Rename the default 'Collection' to MAIN

Create a new collection inside it with the name MAIN_0101 and Move the objects into it.

Rename the plane object to CLOCK:0:20

Tip: CLOCK uses the texture clock.ace by default, it can be changed by using:
CLOCK:n:fontsize:myfont

Tip: MAIN_XXXX = LOD Distance in meters, the 3D cap becomes invisible after 101 m

Look at:

Decoding CabViewControls 1.5 on page xxx for a list of controls, states and data.

THE 3D CAB .CVF FILE

```
1 SIMISA@@@@@@@@@JINX0h0t_____
2
3 Tr_CabViewFile (
4   Position ( 0.0 0.0 0.0 )
5   Direction ( 13 -17 0 )
6   EngineData ( My 3D cab )
7   CabViewControls ( 1
8
9       DigitalClock (
10          Type ( CLOCK DIGITAL_CLOCK )
11          Position (1 1 1 1 )
12          Style ( 24HOUR )
13          Accuracy ( -1 )
14          Justification ( 5 )
15          ControlColour ( 255 255 255 )
16       ) COMMENT( CLOCK:0:18 )
17
18   )
19
20 )
```

Create a cvf file for the 3D cab - or copy the 2D cabs.

- If the train already has a 2D cab, the name of the 3D cab cvf must be the same or the 2D cab is not functional - since the ENG file only has one entry.

Edit the text so it contains the following .

```
SIMISA@@@@@@@@@JINX0h0t_____
Tr_CabViewFile (
  Position ( 0.0 0.0 0.0 )
  Direction ( 0 0 0 )
  EngineData ( My3Dcab )
  CabViewControls ( 1
    DigitalClock (
      Type ( CLOCK DIGITAL_CLOCK )
      Position (1 1 1 1 )
      Style ( 24HOUR )
      Accuracy ( -1 )
      Justification ( 5 )
      ControlColour ( 255 255 255 )
    )
  )
)
```

```
594
595 CabView ( "My3Dcab.cvf" )
596 HeadOut ( -2 3 7 )
597
598 AirBrakesAirCompressorPowerRating( 1.6 )
```

The .ENG file has the entry so be sure the filenames match.

THE 3D CAB & THE ENG FILE

```
569
570
571
572   ORTS3DCab(
573       ORTS3DCabFile ( Cab.s )
574       ORTS3DCabHeadPos ( 0.655 2.2 5.025 )
575       RotationLimit ( 40 60 0 )
576       StartDirection ( 0 0 0 )
577       Sound ( "Vectron_eng.sms")
578   )
579
580 )
581 Engine ( DSB Vectron
582     Wagon ( DSB Vectron )
583     Type ( Electric )
584     MaxPower ( 6400kW )
585     MaxForce ( 350kN )
586     MaxContinuousForce ( 300kN )
587     RunUpTimeToMaxForce ( 4.0 )
588     MaxVelocity ( 200km/h )
589     MaxCurrent ( 1640A )
590     WheelRadius ( 0.62m )
591     MaxSandingTime( -1s )
592     Sanding ( 30km/h )
593     NumWheels ( 1 )
594
595     CabView ( "My3Dcab.cvf" )
596     HeadOut ( -2 3 7 )
597
598     AirBrakesAirCompressorPowerRating( 1.6 )
```

Open the ENG file of the loco you're decided to create a 3D cab for and add the following in the wagon section:

```
ORTS3DCab(
    ORTS3DCabFile ( MyCab.s )
    ORTS3DCabHeadPos ( 0.655 2.2 5.025 )
    RotationLimit ( 40 60 0 )
    StartDirection ( 0 0 0 )
    Sound ( "Mytraincab_eng.sms")
)
```

CabFile (MyCab.s)

The exported shape file.

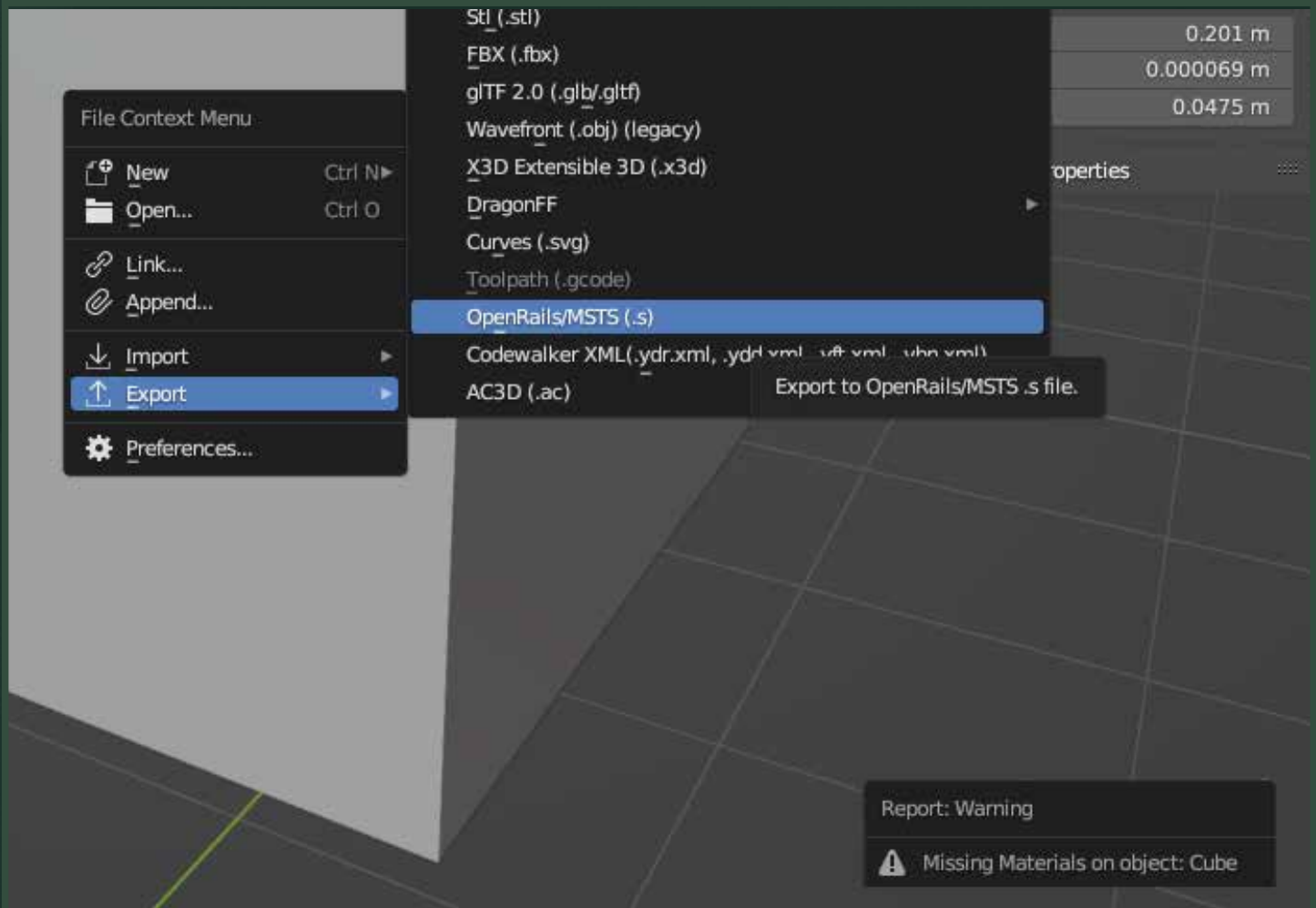
CabHeadPos (0.0 2.4 4.0)

Camera position - 2.2 m up and 5 meter forward from the locos center point

Direction (14.5 0 0)

- up / + down - looking 14.5° down

EXPORT



Exporter warning

Create a materiel texture for the cube to remove it.

THE FOLDER STRUCTURE

Train

- .eng** - The train definition
- .s** - The trains shape file/ Freight animation
- .ace/dds** - The trains texture files

Cabview

- .cvf** - The trains 2D Cab definition
- .ace/dds** - The 2D cabs texture files

3Dcabview

- .cvf** - The trains 3D Cab definition
- .s** - The 3D Cab shape file
- .ace/dds** - The 3D Cab texture files